

Vector Search Libraries

FAISS, Pinecone, Annoy

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Chapter 1. FAISS (Facebook AI Similarity Search)

Open-source library from Meta for efficient similarity search and clustering of dense vectors.

- Runs locally (in-process), no server required
- Supports exact and approximate search (Flat, IVF, HNSW, PQ indexes)
- GPU acceleration available
- No built-in persistence, metadata filtering, or distributed scaling out of the box

Chapter 2. Pinecone

Fully managed, cloud-native vector database (SaaS).

- Zero infra management — API-based
- Built-in metadata filtering, namespaces, hybrid search
- Auto-scaling, real-time upserts/deletes
- Paid service (usage-based pricing)

Chapter 3. Annoy (Approximate Nearest Neighbors Oh Yeah)

Lightweight open-source library from Spotify, built for read-heavy, static datasets.

- Uses random projection trees for ANN search
- Extremely memory-efficient (mmap-based, index can be shared across processes)
- Static index — no easy incremental updates after build
- Simple API, minimal dependencies

Chapter 4. Comparison

Aspect	FAISS	Pinecone	Annoy
Deployment	Self-hosted / embedded	Fully managed cloud	Self-hosted / embedded
Scalability	Manual (sharding, GPU)	Automatic, horizontal	Limited, single-node
Updates	Supports add/remove (index-dependent)	Real-time upsert/delete	Rebuild required for updates
Metadata filtering	Not native (manual workaround)	Native support	Not supported
Persistence	Manual save/load	Managed automatically	mmap file, easy to load
Cost	Free (compute cost only)	Paid (usage-based)	Free
Best index types	Flat, IVF, HNSW, PQ	Proprietary (abstracted)	Random projection trees
Setup complexity	Medium	Low (API-driven)	Low

Chapter 5. When to Use Which

FAISS

Best for research, prototyping, or production systems where you control infra, need GPU speed, or want full control over indexing strategy. Ideal for offline batch search or embedding-heavy pipelines (e.g., RAG prototypes, academic benchmarking).

Pinecone

Best when you want a production-ready, scalable vector DB without managing infra — especially when you need real-time updates, metadata filtering, multi-tenant namespaces, or are shipping a customer-facing RAG application quickly.

Annoy

Best for read-only, memory-constrained scenarios where the dataset doesn't change often — e.g., recommendation systems, static embedding lookups, or lightweight on-device/edge search where simplicity matters more than flexibility.



For your RAG teaching module, FAISS is the natural fit for local notebook demos, while Pinecone is a good real-world contrast point when discussing production-scale vector databases.